The impact of Brexit on the London Stock Exchange: Why did some firms do worse than others?

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May 3, 2022

Abstract

This paper attempts to measure the impact of Brexit on the UK economy by analysing the stock returns of firms listed on the London Stock Exchange. It uses a two-stage estimation process to calculate the abnormal returns following three Brexit events, and then attempts to explain these using firm-specific characteristics. The news of Brexit is found to have had a much larger effect on stock returns than "Brexit Day" itself. Although the majority of firms experienced negative abnormal returns following the outcome of the referendum, this was not a homogeneous result. This heterogeneity can be partially explained by a firm's sector and by its subsidiary structure, with firms more exposed to the UK and EU doing worse. However, the nationality of a firm's board of directors seems to have little to no impact on its abnormal returns. A firm's subsidiary structure remains significant in explaining the size of its cumulative abnormal returns for at least 5 weeks after the result of the referendum was announced.

School of Economics
Economics Dissertation (ECON3001)
Word Count: 7492 words
Supervisor: Rahul Mukherjee

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1 Introduction

On the 23rd June 2016, the population of the United Kingdom voted to leave the European Union. The outcome of the referendum came as a surprise to most people: on the day, Paddy Power had odds of 1/12 and 7/1 for "Remain" and "Leave" respectively (New Statesman (2016)). In the first two trading days following the result of the referendum, global stock markets lost $3 trillion (Bullock (2016)), with the FTSE All-Share Index falling by 7% in this period.

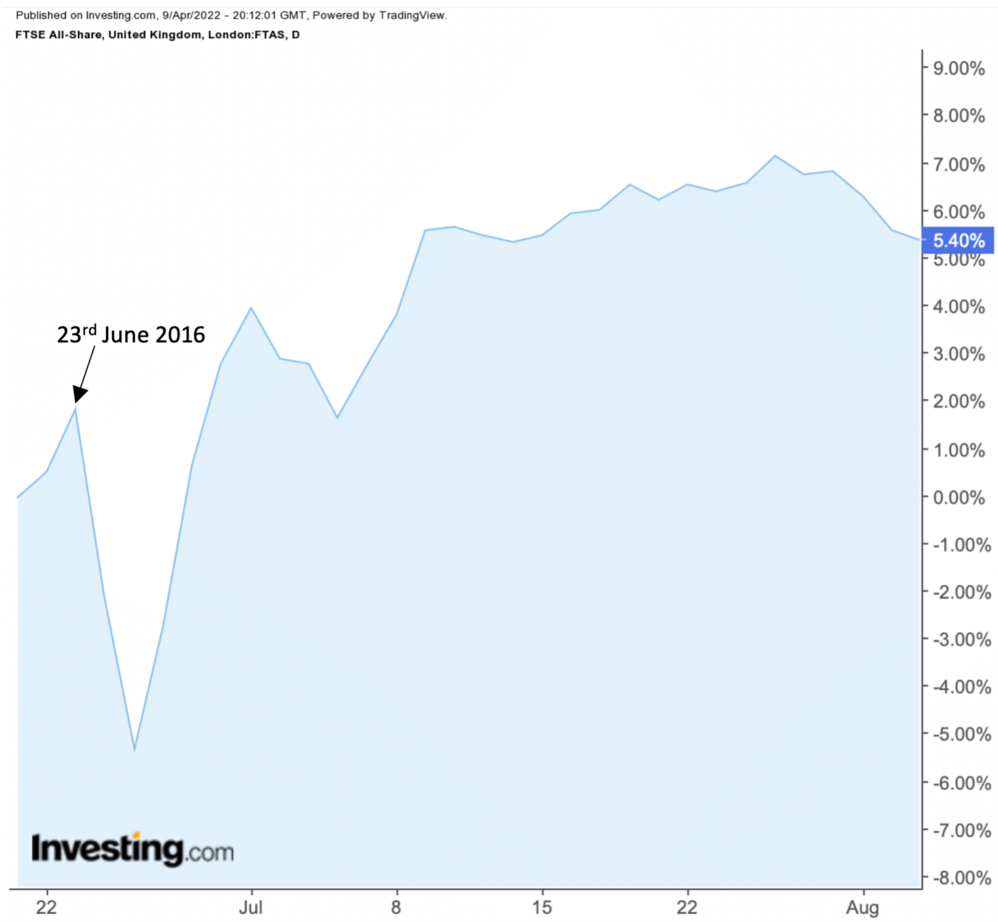

Figure 1: Daily Prices of the FTSE All-Share Index

The FTSE All-Share Index comprises of around 600 of the largest companies that trade on the London Stock Exchange and represents almost 99% of UK market capitalisation. It is the aggregation of the FTSE 100, FTSE 250 and FTSE Small-Cap Indexes.

The European Union describe themselves as "a unique economic and political union between 27 European countries". Its predecessor, the European Economic Community, was founded in 1958 in an attempt to preserve peace in Europe following World War II. It originally comprised of six countries and has since been joined by 22 more countries. The United Kingdom was the first and so far only sovereign country to leave the Union. The EU has a large single market which allows for the free movement of goods, services, capital and people within its member states. It also launched a single European currency, the Euro, which is now used by over 340 million EU citizens, although not all member countries have adopted the currency. In 2015, the European Union comprised of 28 member states (including the UK) and according to the World Bank, had a combined GDP of $13.55 trillion, second only to the United States.

The suggestion of Brexit was originally hinted at back in January 2013, when the then Prime Minister David Cameron declared he was in favour of an in-out EU referendum in the future. Then, the Conservative Party Manifesto for the 2015 General Election pledged "Real change in our relationship with the European Union" and it promised "an in-out referendum by the end of 2017". Following the outcome of the referendum, the UK was originally supposed to withdraw from the EU on the 29th March 2019. However, after years of extensions and negotiations between representatives of the UK and the EU, a Withdrawal Agreement was finally approved by the EU on the 30th January 2020, and the UK left the EU and entered a transition period at 11pm GMT on 31st January 2020 ("Brexit Day"). Then, at 11pm GMT on 31st December 2020, this transition period ended, and the UK left the EU single market and customs union – a "hard Brexit". The overall implications of Brexit have long been a hot topic of discussion amongst economists and politicians and even today, 6 years after the referendum, the long-run effects are still rather ambiguous.

The City of London was widely considered to be the financial capital of Europe: around 35% of EU wholesale financial services activities and 59% of international insurance premiums took place in London pre-Brexit, contributing to a £73 billion trade surplus for the UK in 2014 (Agnew & Jenkins (2016)). Banking and Finance is the UK's largest sector and is the largest contributor to the UK economy. However, only three days after the result of the referendum was announced, banks had already started to take action to shift their European operations out of the UK to emerging financial hotspots such as Dublin, Paris and Frankfurt (Arnold & Noonan (2016)). EU membership gave the banking sector access to skilled EU migrant workers (around 11% of workers in London came from the EU in 2016) and it gave banks a "free passport" to sell in the EU's single market.

There have been various studies using different methods to measure the impact that

Brexit has had and will still have on the UK economy: Born et al. (2017) found that a combination of forward-looking household spending and of lower business spending in response to the vote caused an output loss of more than 1%, while Breinlich et al. (2017) suggested that UK inflation rose by 1.7% within a year due to the referendum. However, this paper will add to the existing literature that considers the stock prices of UK firms and uses the event study methodology to analyse the impact of Brexit. In theory, assuming the semi-strong efficiency of financial markets, share prices should reflect all publicly available information at any given point in time. Investors base their present investment decisions on their expectations of the return of an asset, which is then reflected by that asset's price. Analysing changes in stock prices is therefore a useful way to understand the general feeling of investors towards the potential implications of Brexit overall and its impact on firm-specific performance. Further, calculating abnormal returns[1] is a useful way to see whether a firm or sector, did better or worse than these original expectations.

This paper will examine the stock market reaction of the FTSE All-Share Index following three Brexit events using a two-stage estimation process, similar to Breinlich et al. (2017) and Davies & Studnicka (2018). The three events considered will be the following: 24th June 2016 (the first trading day since the news of Brexit was announced), 21st March 2019 (the day the first Brexit extension was granted by the EU, meaning the UK would no longer leave the EU on 29th March 2019), and 30th January 2020 (the day the Withdrawal Agreement was ratified by the EU and the day before "Brexit Day"). The announcement of the result is found to be far more significant than the other two events, yielding a much larger volume of statistically significant abnormal returns. These abnormal returns can then be partially explained by a firm's sector and its subsidiary structure, with firms more exposed to both the UK and EU seeing larger negative abnormal returns. This remains significant for at least 5 weeks after the outcome of the referendum. However, the nationality of a firm's directors has little explanatory power in regard to its abnormal returns.

[1]Abnormal returns are the difference between a stock's actual returns and its expected returns

2 Literature Review

As mentioned previously, the overall impact of Brexit on the UK is still a widely debated and heavily researched topic in the world of Economics. Therefore, alongside this ever-growing research on the implications of Brexit, this paper adds to the emerging literature on the impact of Brexit on the UK economy and further adds to the existing literature on similar topics. This paper measures the effect of Brexit by looking at stock returns in UK financial markets. Fama (1965) defines an efficient financial market as a market where security prices always fully reflect all available information. Thus, the price of a stock will incorporate expectations and if the market is efficient, one would expect investors to swiftly shift their expectations after an unexpected shock, causing security prices to adjust to a new equilibrium. Peterson (1989) provides a review of event study methodology: a popular analytical tool used in financial research which measures whether a firm or group of firms experience abnormal returns proceeding an event. If markets are efficient, an unexpected shock should only yield abnormal returns for a short amount of time and an expected event, in theory, shouldn't yield abnormal returns. This methodology will be discussed further on in the paper.

The UK's withdrawal from the EU's single market and customs union is a clear barrier to trade. Studies have found a causal relationship between freer trade and increased financial market efficiency. To overcome the problem of variation in the financial market structures of different countries, Baig et al. (2020) collected a sample of American Depositary Receipts (ADRs) which also represents the shares of foreign firms, allowing for variation in trade policies, whilst holding the market structure constant. After conducting a series of multivariate tests, they found a negative relationship between the Fraser Institute's free trade index and two measures of market efficiency: price delay and pricing error. To account for any possible endogeneity in the form of reverse causality, they further supported this by analysing three separate events: two events which reduced trade barriers (Russia joining the WTO and the Canada-European Free Trade Association Free Trade Agreement) and several tweets from former President of the U.S., Donald Trump, which implied the introduction of tariffs on China. They used a difference-in-difference calculation, allowing them to make a causal inference: they concluded that these events supported the hypothesis that freer trade causes increased market efficiency. Thus, in the context of Brexit (a barrier to trade), we would expect UK financial markets to become less efficient. This is supported by Arshad et al. (2019) who examined the volatility and efficiency of the London Stock Exchange around various Brexit vote events. Interestingly, they found that the overall volatility of British financial markets decreased over these events. However, this was not a homogeneous result as various sectors (such as Banking) experienced increased volatility. What was clear on the other hand, was that UK financial markets were much less efficient amid the uncertainty brought about by the Brexit vote, which they attributed to information asymmetry, political uncertainty and liquidity concerns.

Moreover, not only can political uncertainty contribute to the inefficiency of financial markets, it can also directly impact stock prices. Pastor & Veronesi (2012) analysed how changes in government policy impacted stock prices; in particular when there is much uncertainty about these policies. They developed a simple asset pricing model where the government has a decision-making role to account for the endogeneity caused by policy changes not being exogenous. They found that positive returns following an announcement tend to be small since they usually occur when the policy change is generally anticipated, whereas negative returns following an announcement tend to be larger because there is a greater element of surprise. They prove analytically that following the announcement of a policy change, stock prices fall on average. They define two types of uncertainty: impact uncertainty, which concerns uncertainty about the potential implications of the policy change, and political uncertainty, which concerns uncertainty about whether the policy change will take place. They highlight that both types of uncertainty cause the magnitude of the fall in stock prices following the announcement to be larger. In the context of Brexit, most people did not know what the outcome of the referendum would be (political uncertainty) and the implications of Brexit were extremely unclear and still to this day, the long-run impact of Brexit is not fully known (impact uncertainty). Hence, applying the findings of Pastor & Veronesi (2012) to the Brexit vote, stock prices should fall and due to the large amount of uncertainty surrounding Brexit, they should fall by a significant amount (which was the case since the FTSE All-Share Index fell 7% in the two days following the referendum).

Furthermore, Brexit did not have a homogeneous impact on all firms in the UK and in fact, a lot of the heterogeneity observed in the stock market can be attributed to a specific firm's sector. Ramiah et al. (2016) analysed the impact of Brexit, relative to expectations, by using the event study methodology to calculate the abnormal returns (ARs) and cumulative abnormal returns (CARs) across various sectors. They found that 11 industries reacted as expected prior to the event, whereas 13 industries reacted in an unexpected manner. For example, the Banking sector reacted in a way that was expected, experiencing statistically significant ARs of -4.99% on the day of the event, while the Aerospace and Defence sector reacted in an unexpected manner, experiencing statistically significant positive ARs of 2.09% on the day of the event. Their results highlight the sectoral heterogeneity of the stock market reaction to the Brexit vote. They concluded that although this initial reaction does not necessarily highlight the long-term impact of Brexit, it certainly gives an indication of what the long-term implications may be.

Breinlich et al. (2018) and Davies & Studnicka (2018) both use a two-stage estimation process to estimate the impact of Brexit on firms listed on the London Stock Exchange. They first measure the abnormal returns and cumulative abnormal returns of these firms, and then try to explain the ARs and CARs by using them as the dependent variable in a second regression. Breinlich et al. (2018) use a wide range of explanatory variables and link these to forecasts made before the Brexit vote. They used the event study methodology to estimate the ARs of around 350 companies. They found that the stock market reacted negatively to the announcement, mainly due to the depreciation of the pound and the expectation of an economic downturn. Further, they found that the share prices of companies more exposed to the UK market reacted more severely to the referendum result. Davies & Studnicka (2018) use a firm-by-firm OLS approach of the FTSE 350 to estimate the ARs and CARs of each firm following 6 Brexit "events". They also regress their AR findings with a number of explanatory variables, with a focus on global value chains (GVCs). They find a significant amount of heterogeneity in the returns of firms in the FTSE 350. From their second regression, using the abnormal returns as the dependent variable, they highlight that this can be explained by a firm's GVC structure. They found that firms with a larger share of affiliates in the UK or the EU seemed to have larger negative abnormal returns, but that this negative effect was more severe for firms with a greater share of UK affiliates. They further show that the location of a firm's affiliates can explain the abnormal returns of a firm for up to a month after the Brexit announcement.

In summary, this paper adds to the existing literature on the impact of Brexit. It uses event study methodology to calculate the abnormal returns and cumulative abnormal returns of individual firms and their sectors. It then uses a two-stage estimation similar to the one used by Davies & Studnicka (2018) but builds on this paper in three ways: firstly, it uses a larger sample size as it looks at firms in the FTSE All-Share index; secondly, it considers other explanatory variables to explain the abnormal returns (specifically, whether the nationality of a firm's directors has an effect on the abnormal returns); and finally, it considers two events that occurred after the Davies & Studnicka (2018) paper was published.

3 Data & Methodology

This paper uses a two-stage estimation process by combining event study methodology and an OLS regression to estimate the impact of Brexit on firms listed on the London Stock Exchange, relative to expectations.

3.1 Event Study Methodology

Fama et al. (1969) originally introduced the event study methodology most commonly used today when they analysed the relationship between stock splits and dividend increases. Peterson (1989) provides a review of event study methodology, and it is now a popular analytical tool in financial research.

3.1.1 Estimation and Event Window

Event studies aim to calculate the abnormal returns of a firm following a specific event. Abnormal returns are the difference between a firm's actual returns relative to its expected returns. This can occur when investors shift their expectations following additional information, causing a firm's actual return to differ from its expected return. When conducting an event study, one must first define an estimation window and an event window.

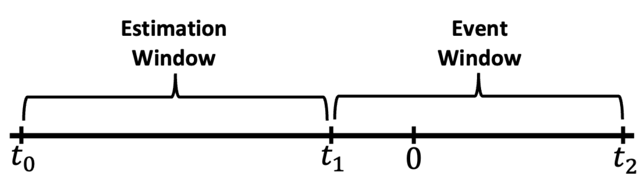

Figure 2: Event Studies Timeline

Figure 2 shows the timeline of an event study, where the event occurs at time $t = 0$. The estimation window is the period t_0 to t_1, where t_1 occurs just before the event. The estimation window is used to calculate the normal returns. In this study, t_1 occurs 15 trading days before the various events. Peterson (1989) suggests an estimation window of between 100 to 300 days, where a longer estimation window should lead to more accurate results. The event window is the period t_1 to t_2, which is used to calculate the abnormal returns. For Event 1 (24th June 2016), the estimation window is 200 days long and the event window finishes 25 days after the event. For Event 2 (21st March 2019), the estimation window is slightly shorter so that it doesn't overlap with many other Brexit events[2] and is 150 days, with an event window that also finishes 25 days after the event. For Event 3 (30th January 2020), the estimation window is 150 days, but the event window finishes 15 days after the event[3].

3.1.2 Stock Data Collection

Once the estimation and event windows have been established, the stock data must be obtained for the individual firms and the market index. This paper uses the firms that make up the FTSE All-Share Index, which comprises of around 600 companies. The list of these firms was obtained from the Bureau van Dijk FAME database which has data on over 11 million British and Irish firms. There were a total of 569 active firms in the FTSE All-Share listed on the FAME database. The reason the list of firms was taken from the FAME database, rather than somewhere else, is because this allowed me to obtain firm-specific data and controls for the second stage of the estimation process, which will be discussed later.

The historical stock prices of these 569 firms were then collected from Yahoo Finance and the historical price of the FTSE All-Share Index was taken from Investing.com, since this was more easily accessible than Yahoo Finance. Having obtained the historical prices of the firms and FTSE All-Share Index over the estimation and event windows for all three events, the daily log-returns were then calculated as follows:

$$R_{i,t} = \ln(\tfrac{P_{i,t}}{P_{i,t-1}}) = \ln(P_{i,t}) - \ln(P_{i,t-1})$$

where $R_{i,t}$ is firm i's daily stock return, $P_{i,t}$ is the closing price of firm i's stock at time t, and $P_{i,t-1}$ is the closing price of firm i's stock from the previous day. The daily log-return is equal to the percentage change of the price of a stock from day $t-1$ to day t. Since some of the firms were founded or became publicly listed after the three events studied in this paper, and thus had no available historical prices for those dates, not all 569 firms were included – event 1 includes 488 firms, event 2 includes 543 firms and event 3 includes 546 firms.

[2]Davies & Studnicka (2018) conclude that this has no qualitative and only a small quantitative impact on their results

[3]This is because towards the end of February 2020, the stock market started to crash due to Covid-19 which would have significantly skewed our results

3.1.3 Abnormal Returns

Having obtained the daily log-returns for the estimation and event windows, the abnormal returns could then be calculated. Firstly, normal returns of each firm can be calculated from the estimation window by using the OLS returns model suggested by Fama et al. (1969):

$$R_{i,t} = \alpha_i + \beta_i R_{FTSEAllShare,t} + \varepsilon_{i,t}, \qquad t_0 \leq t \leq t_1$$
$$E[\varepsilon_{i,t}] = 0, \quad V[\varepsilon_{i,t}] = \sigma^2_{\varepsilon_i}$$

where $R_{i,t}$ is firm i's daily stock return, α_i is the intercept, β_i is the beta (a measure of risk) of stock i, $R_{FTSEAllShare,t}$ is the return of the FTSE All-Share index, and $\varepsilon_{i,t}$ is an error term with mean zero.

The parameters α_i and β_i can then be used to estimate the expected returns of each firm i in the event window. The abnormal returns (ARs) of firm i is then equal to its actual returns during the event window, minus its expected returns:

$$E[R_{i,t}] = \alpha_i - \beta_i R_{FTSEAllShare,t}, \qquad t_1 \leq t \leq t_2$$
$$AR_{i,t} = R_{i,t} - E[R_{i,t}]$$

When we are trying to calculate the abnormal returns over a period longer than one day, a time-series aggregation of the ARs becomes necessary. This gives rise to the cumulative abnormal returns (CARs):

$$CAR_i(T_1, T_2) = \textstyle\sum_{t=T_1}^{T_2} AR_{i,t}$$

where $t_1 \leq T_1 < T_2 \leq t_2$. Studying the CARs over a period of days rather than the separate ARs for those days helps to account for overshooting in a firm's daily return – if a firm has a positive AR one day but a negative AR the next, the two will cancel out if you look at the CAR over those two days.

In order to examine the impact of Brexit on various sectors, the abnormal returns of the firms in each sector j need to be aggregated and averaged, giving us the average abnormal returns (AARs):

$$AAR_{j,t} = \tfrac{1}{N_j} \textstyle\sum_{i=1}^{N_j} AR_{i,t}$$

where N_j is the number of firms in sector j ($1 \leq j \leq 24$). Again, if we want to study the average effect over a period longer than one day, a time-series aggregation of the AARs is necessary. This gives rise to the cumulative average abnormal returns:

$$CAAR_j(T_1, T_2) = \sum_{t=T_1}^{T_2} AAR_{j,t}$$

where $t_1 \leq T_1 < T_2 \leq t_2$.

3.1.4 Significance Tests

When conducting event studies, we usually want to test whether the ARs or CARs are significantly different from zero (in a statistical sense). That is, we want to test the following:

$$H_0 : (C)(A)AR_i = 0$$
$$H_A : (C)(A)AR_i \neq 0$$

However, a standard t-test suffers from the issue of event-date clustering. This leads to two problems: event-induced volatility changes and cross-sectional correlation of ARs, which can both significantly distort our results. Event-induced volatility changes often occur when events are clustered but cross-sectional correlation of ARs occurs when the focus is on a single event which affects multiple firms at the same time. Therefore, this study is likely to suffer from both issues and so a test statistic which accounts for this is needed. Boehmer et al. (1991) looked at the impact of event-induced variance and found that even a small change in the variance leads to a disproportionate increase in Type I errors[4] for the most commonly-used methods. They proposed a test statistic which is robust against any event-induced variance using a standardised cross-sectional method. To test $H_0 : (A)AR_i = 0$, the BMP test statistic at day t is given by:

$$z_{BMP,t} = \frac{ASAR_t}{\sqrt{N}.\hat{\sigma}_{ASAR_t}}$$

where $ASAR_t$ is the sum of standardised abnormal returns over the sample ($ASAR_t = \sum_{i=1}^{N} SAR_{i,t} = \sum_{i=1}^{N} \frac{AR_{i,t}}{\hat{\sigma}_{AR_{i,t}}}$) and the standard deviation is given by:

$$\hat{\sigma}^2_{ASAR_t} = \frac{1}{N-1} \sum_{i=1}^{N} (SAR_{i,t} - \frac{1}{N} \sum_{j=1}^{N} SAR_{j,t})^2$$

Similarly, if we are trying to test $H_0 : C(A)AR_i = 0$, the BMP test statistic is given by:

$$z_{BMP} = \sqrt{N} \frac{\overline{SCAR}}{\hat{\sigma}_{\overline{SCAR}}}$$

where $\overline{SCAR}$ is the average standardised cumulative abnormal returns across the N firms. Its standard deviation is given by:

[4] A Type I error is when you incorrectly reject the null hypothesis when it is true

$$\hat{\sigma}_{\overline{SCAR}} = \frac{1}{N-1}\sum_{i=1}^{N}(SCAR_i - \overline{SCAR})^2,$$

$$\overline{SCAR} = \frac{1}{N}\sum_{i=1}^{N} SCAR_i = \frac{1}{N}\sum_{i=1}^{N} \frac{CAR_i}{\hat{\sigma}_{CAR_i}}$$

where $\hat{\sigma}_{CAR_i}$ is the forecast-error-corrected standard deviation from Mikkelson & Partch (1988).

This paper uses the adjusted-BMP test proposed by Kolari & Pynnönen (2010) which also accounts for the cross-sectional correlation of ARs. They define $\overline{r}$ to be the average of the sample cross-correlations of the estimation-period abnormal returns. To test $H_0 : (A)AR_i = 0$, the adjusted-BMP test statistic at day t is:

$$z_{BMP,t} = z_{BMP,t}\sqrt{\frac{1-\overline{r}}{1+(N-1)\overline{r}}}$$

We can see that if $\overline{r} = 0$, the adjusted-BMP test statistic becomes the same as the original BMP test statistic. This test statistic can also be used when we consider cumulative abnormal returns, if we assume the square-root rule holds for the standard deviation of different periods. Thus, if this assumption holds, to test $H_0 : C(A)AR_i = 0$, the adjusted-BMP test statistic is again:

$$z_{BMP,t} = z_{BMP,t}\sqrt{\frac{1-\overline{r}}{1+(N-1)\overline{r}}}$$

3.2 Firm Characteristics

The second stage of the estimation process attempts to explain the ARs and CARs using firm-specific characteristics. Davies & Studnicka (2018) have a particular focus on Global Value Chain (GVC) structure when selecting their firm-specific characteristics. This is due to the findings of Hanson et al. (2005) who highlight the importance of intra-firm trade in multinational GVCs. Thus, Davies & Studnicka (2018) use the location of a multinational's affiliates as a measure of GVC activity. This paper replicates this approach and obtains information on firms' subsidiary locations – the number of subsidiaries and the individual subsidiary country ISO codes were collected from the FAME database. The number of subsidiaries located in the UK, as well as in the rest of the 27 EU countries, were calculated in order to work out the share of UK subsidiaries and EU subsidiaries. Firms that had no subsidiaries were excluded from the sample. A summary of this data is given in Table 1.

$$\text{Share of UK Subsidiaries} = \frac{\text{Number of UK Subsidiaries}}{\text{Total Number of Subsidiaries}}$$

$$\text{Share of EU Subsidiaries} = \frac{\text{Number of EU Subsidiaries}}{\text{Total Number of Subsidiaries}}$$

Subsidiaries	Observations	Mean	St. Dev.	Min	Max
Total	569	100.5	164.8	0	1606
No. UK	569	41.0	73.9	0	841
No. EU	569	14.8	34.4	0	365
Share UK	546	50.3%	36.7%	0%	100%
Share EU	546	13.3%	18.4%	0%	100%

Table 1: Subsidiary Summary Statistics

As well as considering a firm's subsidiary structure when trying to explain the ARs and CARs, this paper also has a focus on the firm's board of directors. Specifically, it looks at the nationalities of these directors at the time of each event. The number of directors and their respective nationalities, along with their appointment and resignation dates, were taken from the FAME database to work out the share of British directors and the share of European (EU-27 countries) directors at the time of each event. A summary of this data is given in Table 2.

$$\text{Share of British Directors} = \frac{\text{Number of British Directors}}{\text{Total Number of Directors}}$$

$$\text{Share of European Directors} = \frac{\text{Number of European Directors}}{\text{Total Number of Directors}}$$

Directors	Observations	Mean	St. Dev.	Min	Max
Event 1					
Total	454	6.3	2.6	1	18
No. British	454	5.1	2.2	0	14
No. European	454	0.4	0.9	0	7
Share British	454	84.4%	22.0%	0%	100%
Share European	454	5.6%	10.7%	0%	58.3%
Event 2					
Total	491	6.3	2.6	1	16
No. British	491	4.9	2.2	0	16
No. European	491	0.5	0.9	0	6
Share British	491	81.4%	23.9%	0%	100%
Share European	491	6.7%	12.2%	0%	60%
Event 3					
Total	497	6.3	2.5	1	16
No. British	497	4.9	2.1	0	16
No. European	497	0.5	0.9	0	6
Share British	497	80.7%	23.7%	0%	100%
Share European	497	6.6%	11.7%	0%	60%

Table 2: Director Summary Statistics

This paper attempts to explain these ARs and CARs by running an OLS regression with the abnormal and cumulative abnormal returns as the dependent variable. The share of UK and EU subsidiaries, along with the share of British and European directors, are used as the independent variables. Three control variables are also included: the log of a firm's market capitalisation from a month before the event (also taken from the FAME database), the log of the number of subsidiaries and the log of the number of directors. This controls for the size of the firms. Although it would have been preferable to control for the size of each subsidiary, missing data from the FAME database meant this was not feasible. Each firm's sector was also obtained from the FAME database, so a vector of sector dummies is also included in the regression to account for the fact that individual sectors reacted differently to Brexit (Ramiah et al. (2016)). The final regression is therefore given by:

$$AR_i/CAR_i = \beta_1 \text{ShareUKSubs}_i + \beta_2 \text{ShareEUSubs}_i + \beta_3 \text{ShareUKDir}_i + \beta_4 \text{ShareEUDir}_i + \\ \beta_5 \ln(\text{MktCap})_i + \beta_6 \ln(\text{NumSubs})_i + \beta_7 \ln(\text{NumDir})_i + \alpha_S + \varepsilon_i$$

where α_S is the aforementioned vector of sector dummies and ε_i is an error term with mean zero. Finally, the dependent variable has been constructed and thus has the potential to introduce heteroskedasticity (Lewis & Linzer (2005)). To account for this, the White robust standard error correction is used.

3.3 Hypotheses

3.3.1 First Stage

The first stage of the estimation process calculates the abnormal returns of each firm. When calculating abnormal returns, we would like to test whether they are statistically significantly different from zero. As explained above, a standard t-test faces the issue of event clustering leading to event-induced volatility changes and cross-sectional correlation, so the adjusted-BMP test is used in this paper. To start with, we would like to test whether the abnormal returns on the day of the event are significant, giving us our first hypothesis:

$$H_0 : AR_i = 0$$
$$H_A : AR_i \neq 0$$

This is done on a firm-by-firm approach. We would then like to look at the longevity of these abnormal returns and so the cumulative abnormal returns are calculated. We can then test to see whether these significantly differ from zero, giving us our second hypothesis:

$$H_0 : CAR_i = 0$$
$$H_A : CAR_i \neq 0$$

3.3.2 Second Stage

The second stage of the estimation process attempts to explain these abnormal returns by using an OLS regression with the abnormal returns as the dependent variable and a number of firm-specific characteristics as the independent variables. We would like to test whether these firm characteristics are statistically significant in explaining the size of a firm's abnormal returns. The first three hypotheses relate to a firm's subsidiary structure. We would like to test whether the share of a firm's subsidiaries located in the UK plays a role in explaining the size of its abnormal returns. Thus, the first hypothesis is:

$$H_0 : \beta_1 = 0$$
$$H_1 : \beta_1 \neq 0$$

Similarly, we would like to test whether the share of a firm's subsidiaries located in the EU helps to explain its abnormal returns. Hence, our second hypothesis is:

$$H_0 : \beta_2 = 0$$
$$H_2 : \beta_2 \neq 0$$

If one or both of these null hypotheses are rejected, we can conclude that a firm's GVC structure helps to explain the size of its abnormal returns. We would also like to test whether one of these has a larger effect, giving us our third hypothesis:

$$H_0 : \beta_1 - \beta_2 = 0$$
$$H_3 : \beta_1 - \beta_2 \neq 0$$

If the null hypothesis is rejected, we can conclude that either the share of UK subsidiaries or the share of EU subsidiaries has a greater effect on the size of a firm's abnormal returns.

This paper also tries to relate the size of a firm's abnormal returns to the nationalities of its board of directors. We would like to test whether the share of British directors on a firm's board of directors has an impact on its abnormal returns. Therefore, our fourth hypothesis is:

$$H_0 : \beta_3 = 0$$
$$H_4 : \beta_3 \neq 0$$

Similarly, we would like to test whether the share of European directors on a firm's board of directors has an impact on its abnormal returns, giving us our fifth hypothesis:

$$H_0 : \beta_4 = 0$$
$$H_5 : \beta_4 \neq 0$$

Again, if one or both of these null hypotheses are rejected, we can conclude that the nationalities of a firm's board of directors helps to explain the size of its abnormal returns. Finally, we would like to test whether one of these has a larger effect, giving us our final hypothesis:

$$H_0 : \beta_3 - \beta_4 = 0$$
$$H_6 : \beta_3 - \beta_4 \neq 0$$

If the null hypothesis is rejected, we can conclude that either the share of British directors or the share of European directors has a greater effect on the size of a firm's abnormal returns.

4 Results & Discussion

4.1 Event 1

4.1.1 First Stage

Window	*Observations*	*Mean (%)*	*St. Dev. (%)*	*Min (%)*	*Max (%)*	*No. Significant*
(-2,-2)	488	0.05	1.61	-8.57	9.59	57
(-1,-1)	488	0.41	1.62	-5.42	10.13	54
(0,0)	488	-3.08	6.85	-31.11	13.50	333
(0,+1)	488	-6.75	12.05	-59.74	22.20	353
(0,+2)	488	-6.15	11.25	-59.25	17.79	332
(0,+3)	488	-5.55	10.96	-51.89	24.33	307
(0,+4)	488	-5.71	11.43	-50.79	26.16	294
(0,+7)	488	-7.62	14.36	-62.23	39.39	299
(0,+9)	488	-7.19	14.94	-62.91	39.98	282
(0,+14)	488	-3.95	12.47	-45.77	38.84	213
(0,+19)	488	-3.03	12.41	-48.82	42.98	173
(0,+24)	488	-2.17	12.81	-51.68	41.66	157

Table 3: Cumulative Abnormal Returns (CARs)

Table 3 depicts the cumulative abnormal returns of 488 firms for up to 25 days after the Brexit referendum result. Since the decision to leave the EU came as a surprise to most people, one would expect a significant number of abnormal returns as investors shift their expectations following the news. In fact, on the first day following the announcement, over two thirds of firms in this sample experienced statistically significant abnormal returns and even after 25 trading days (equivalent to 5 weeks), over 30% of firms had significant cumulative abnormal returns. Table 3 suggests that the average firm's actual return was worse than expected following the result. On the day of the event, the average firm's return was 3.08% lower than expected and the following day, this rose to 3.67% (giving a CAR of -6.75%). Table 3 also highlights the heterogeneity of the FTSE All-Share's response to the Brexit vote. This can be seen by the increasing standard deviation after the event, indicating a large amount of variation of the CARs, and the difference between the minimum and maximum values. For example, the standard deviation of the CARs peaks 10 days after the event at 14.94%, at which point the worst firm's CAR was -62.91% and the best firm's CAR was 39.98%. Moreover, Table 3 shows that the number of significant abnormal returns fell over time which would be expected. According to the Efficient Market Hypothesis (Fama (1965)), one would expect investors to swiftly adjust their expectations following an unexpected shock so that each firm's stock price once again represents all available information. However, the fact that even after 5 weeks, 161 firms experienced statistically significant CARs, perhaps suggests the market is slightly inefficient.

4.1.2 Second Stage

The second stage attempts to relate each firm's CARs with their respective characteristics.

	(1)	(2)	(3)	(4)	(5)
VARIABLES	CAR(0,0)	CAR(0,+1)	CAR(0,+2)	CAR(0,+3)	CAR(0,+4)
Share UK Subs.	-0.0878***	-0.177***	-0.169***	-0.166***	-0.176***
	(0.00829)	(0.0138)	(0.0131)	(0.0132)	(0.0143)
Share EU Subs.	-0.0668***	-0.150***	-0.132***	-0.132***	-0.137***
	(0.0145)	(0.0238)	(0.0228)	(0.0223)	(0.0230)
Share UK Dir.	-0.0378***	-0.0234	-0.0198	-0.0257	-0.0292
	(0.0139)	(0.0263)	(0.0259)	(0.0280)	(0.0310)
Share EU Dir.	-0.0261	-0.0303	-0.0434	-0.0550	-0.0445
	(0.0303)	(0.0530)	(0.0519)	(0.0516)	(0.0558)
Market Capitalisation	-0.00475*	-0.00960**	-0.00479	-0.00584	-0.00592
	(0.00242)	(0.00414)	(0.00403)	(0.00404)	(0.00448)
No. Subsidiaries	0.00109	0.00721	0.00502	0.00291	0.00285
	(0.00258)	(0.00483)	(0.00468)	(0.00490)	(0.00532)
No. Directors	-0.00341	0.00196	-0.000562	0.00132	-0.00105
	(0.00837)	(0.0134)	(0.0137)	(0.0131)	(0.0137)
Constant	0.137***	0.184***	0.126**	0.152***	0.157***
	(0.0306)	(0.0538)	(0.0510)	(0.0522)	(0.0573)
Sector FE	YES	YES	YES	YES	YES
Observations	407	407	407	407	407
R-squared	0.568	0.592	0.562	0.537	0.509

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 4: Size of Cumulative Abnormal Returns (CARs)

Table 4 shows how the firm characteristics mentioned in the methodology section relate to a firm's CARs. Hanson et al. (2005) found that trade barriers significantly reduce intra-firm trade within US multinationals. Thus, one would expect Brexit to reduce the efficiency of a firm's GVC, meaning investors would likely want to sell the shares of firms who are more exposed to the UK and EU, causing their share price to fall. It therefore makes sense that both the share of UK subsidiaries and the share of EU subsidiaries have negative coefficients, with both being significant at the 1% significance level for the first five days of the event window. We would also expect the effect to be greater for the share of UK subsidiaries than

EU subsidiaries, since it is down to the UK to negotiate new trade deals with not only the EU, but also the rest of the world. From Table 4, we can see that this is the case for all 5 days; however, this wasn't statistically significant.

Now looking at the nationality of the board of directors, one can see that the share of British directors and the share of European directors both also have a negative effect on a firm's CARs. This could potentially be due to the increased uncertainty over the living situation of these directors and the free movement of people. For example, after the decision to leave the EU it was unclear whether EU nationals would still legally be allowed to live in the UK or whether British nationals would still be allowed to live in the EU. However, the share of European directors is insignificant for all 5 days, while the share of British directors is only significant on the day and not thereafter. Further, on the day the share of British directors had a greater effect than the share of European directors on a firm's CARs, but from then on, the share of European directors had a larger impact. However, this was statistically insignificant for all 5 days.

VARIABLES	(6) CAR(0,+7)	(7) CAR(0,+9)	(8) CAR(0,+14)	(9) CAR(0,+19)	(10) CAR(0,+24)
Share UK Subs.	-0.229***	-0.251***	-0.209***	-0.197***	-0.201***
	(0.0168)	(0.0175)	(0.0161)	(0.0163)	(0.0180)
Share EU Subs.	-0.166***	-0.163***	-0.120***	-0.0962***	-0.0935***
	(0.0272)	(0.0292)	(0.0276)	(0.0284)	(0.0305)
Share UK Dir.	-0.0209	-0.0161	-0.0175	-0.0301	-0.0212
	(0.0358)	(0.0394)	(0.0342)	(0.0377)	(0.0431)
Share EU Dir.	-0.00185	-0.0195	-0.00220	-0.00507	-0.0250
	(0.0625)	(0.0629)	(0.0542)	(0.0586)	(0.0653)
Market Capitalisation	-0.0106**	-0.00921*	-0.0106**	-0.0143***	-0.0144**
	(0.00517)	(0.00554)	(0.00492)	(0.00501)	(0.00557)
No. Subsidiaries	0.00210	0.000333	0.000580	-0.000327	-0.00283
	(0.00612)	(0.00656)	(0.00596)	(0.00604)	(0.00642)
No. Directors	0.00550	0.00824	0.0113	0.0115	0.0144
	(0.0163)	(0.0173)	(0.0161)	(0.0161)	(0.0166)
Constant	0.218***	0.213***	0.228***	0.295***	0.301***
	(0.0669)	(0.0689)	(0.0569)	(0.0658)	(0.0734)
Sector FE	YES	YES	YES	YES	YES
Observations	407	407	407	407	407
R-squared	0.562	0.573	0.546	0.519	0.474

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 5: Long-Term Size of Cumulative Abnormal Returns (CARs)

Table 5 shows the CARs for up to 25 trading days. One can see that the subsidiary structure of a firm remained significant in explaining the size of its CARs for at least 5 weeks after the result of the referendum was announced. Unlike the first five days of the event window, the share of UK subsidiaries had a statistically significantly greater impact on the CARs of a firm than the share of EU subsidiaries (at the 5% level for the (0,+7) window and at the 1% level thereafter). However, the share of British directors, although having a significantly negative effect on the day of the event, has no power in explaining the CARs of a firm for the following 24 days. Further, the share of European directors appears to have no effect at all on the CARs of a firm for all 25 days.

VARIABLES	(1)	(2)	(3)	(4)	(5)
Share UK Subs.	-0.202***	-0.233***	-0.218***	-0.215***	-0.177***
	(0.0144)	(0.0136)	(0.0135)	(0.0136)	(0.0138)
Share EU Subs.		-0.173***	-0.171***	-0.150***	-0.150***
		(0.0251)	(0.0250)	(0.0254)	(0.0238)
Share UK Dir.			-0.0556**	-0.0869***	-0.0234
			(0.0236)	(0.0274)	(0.0263)
Share EU Dir.				-0.109*	-0.0303
				(0.0558)	(0.0530)
Market Capitalisation	-0.00608	-0.00454	-0.00641	-0.00663	-0.00960**
	(0.00451)	(0.00443)	(0.00451)	(0.00452)	(0.00414)
No. Subsidiaries	-0.00260	-0.00382	-0.00351	-0.00303	0.00721
	(0.00606)	(0.00586)	(0.00574)	(0.00571)	(0.00483)
No. Directors	-0.00405	-0.00555	-0.0125	-0.0131	0.00196
	(0.0132)	(0.0127)	(0.0132)	(0.0131)	(0.0134)
Constant	0.132***	0.154***	0.231***	0.261***	0.184***
	(0.0449)	(0.0438)	(0.0584)	(0.0609)	(0.0538)
Sector FE	NO	NO	NO	NO	YES
Observations	407	407	407	407	407
R-squared	0.368	0.413	0.420	0.425	0.592

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 6: Contribution to R-squared for window CAR(0,+1)

Table 6 shows how much each variable contributes to the R-squared for the window (0,+1). Considering our results above, it is not a surprise that the share of UK subsidiaries and the share of EU subsidiaries contribute far more to the R-squared than the share of British and the share of European directors. What is also interesting, is how much the sector fixed effects variable contributes to the R-squared: increasing it by 0.167. This highlights that a firm's abnormal returns can be partially explained by which sector of the economy it operates in, rather than just its individual characteristics, which supports the findings of Ramiah et al. (2016). Moreover, the sector fixed effects variable is clearly important because column (4), where the fixed effects variable is not included, suggests that both the share of British and the share of European directors are significant in explaining the size of a firm's CARs. Thus, without this variable, one might incorrectly reject the null hypothesis that the nationality of a firm's directors does not impact the size of its CARs.

4.1.3 Summary/Implications

To summarise, the day of the first event saw more than a six-fold increase in the number of statistically significant abnormal returns compared to the previous day, with the average firm seeing negative abnormal returns. The number of significant cumulative abnormal returns remained well above the pre-event level for at least 25 trading days after the event. It is found that the size of these abnormal returns can be explained by a firm's sector and by its exposure to the UK and EU, with firms more exposed to these areas seeing larger negative abnormal returns. Thus, if a country unexpectedly decides to leave a trade union or economic union, one might expect the firms of that country to be more negatively affected if they are more exposed to those regions. Similarly, if a firm wishes to be less exposed to a negative shock like this, they could diversify their operations so that they are more globalised. However, since events like this are not a regular occurrence, it likely does not justify the costs of expanding a firm's operations if the sole reason for it was to minimise the effects of an event like this. That is not to say though that an event like this will not happen again. For example, if Marine Le Pen had recently won the French election, there would likely have been big changes in the relationship between France and the EU. She originally supported a French Brexit-style withdrawal from the EU, but later changed to supporting a radical modification of the trading bloc (Euronews (2020)). Either way, if she introduced a policy which increased trade barriers between France and the EU, the results here would suggest that firms more exposed to France and the EU would be more negatively affected.

4.2 Events 2 & 3

4.2.1 First Stage

Events 2 and 3 were far more expected than the first event and so one would expect the stock market reaction to be far less severe, since an efficient financial market should reflect all available information. This was clearly the case, as seen in Table 7. Just over 10% of firms had significant ARs on the day of the second event, while just over 12% had significant ARs on the day of the third event – far less than almost 70% in the first event. For both events, the number of significant results swiftly shifts back to the level before the event (one day for the second event and two days for the third event), so it seems likely that the CARs after this were not due to the respective events. This suggests that prices have adjusted to any additional information, supporting the Efficient Market Hypothesis (Fama (1965)).

Window	Observations	Mean (%)	St. Dev. (%)	Min (%)	Max (%)	No. Significant
Event 2						
(-2,-2)	543	0.14	1.61	-9.12	6.25	36
(-1,-1)	543	-0.18	1.85	-11.05	12.72	44
(0,0)	543	-0.53	1.84	-12.52	9.75	59
(0,+1)	543	-0.48	2.07	-9.68	7.67	34
(0,+2)	543	-0.96	2.55	-13.23	7.23	35
(0,+3)	543	-1.08	2.94	-13.08	9.78	32
(0,+4)	543	-0.82	3.12	-12.87	14.53	22
(0,+7)	543	-0.52	4.06	-19.11	16.64	25
(0,+9)	543	-0.14	4.22	-20.78	18.15	14
(0,+14)	543	-0.71	8.24	-126.99	24.5	20
(0,+19)	543	0.31	8.58	-97	27.95	27
(0,+24)	543	1.22	9.2	-91.97	40.04	27
Event 3						
(-2,-2)	546	0.01	1.69	-10.14	14.34	33
(-1,-1)	546	0.09	1.51	-8.55	8.34	39
(0,0)	546	0.08	1.77	-13.86	7.82	67
(0,+1)	546	0.32	2.35	-12.69	13.63	54
(0,+2)	546	0.14	2.78	-9.69	19.01	35
(0,+3)	546	0.27	3.47	-34.14	31.79	28
(0,+4)	546	0.36	3.77	-33.3	27.28	38
(0,+7)	546	0.07	4.66	-31.22	28.06	35
(0,+9)	546	0.3	4.98	-29.59	36.24	26
(0,+14)	546	0.53	6.53	-32.55	63.76	32

Table 7: Cumulative Abnormal Returns (CARs)

Table 7 suggests that on average, firms did slightly worse than expected on the day of the second event. Although one could argue that it was a positive thing that the Brexit extension was granted (as it meant there was more time to negotiate with the EU to avoid a "No-Deal Brexit"), it also created more uncertainty about when or even whether the UK would leave the EU. Moreover, after the third event, it seems that firms did slightly better than expected. This is likely because it was a positive thing that the Withdrawal Agreement was ratified by the EU, since it meant there was more clarity about the future post-Brexit.

4.2.2 Second Stage

From Table 8, the share of UK and EU subsidiaries both seem to have a negative effect on a firm's abnormal returns on the day of the second event. The reason for this is likely to be similar to the first event but to less of an extent. The share of British directors has a slight

VARIABLES	(1) CAR(0,0)	(2) CAR(0,+1)	(3) CAR(0,+2)	(4) CAR(0,+3)	(5) CAR(0,+4)	(6) CAR(0,+7)	(7) CAR(0,+9)
Share UK Subs.	-0.0100***	-0.00989***	-0.0122***	-0.0114***	-0.00549	-0.0147**	-0.00829
	-0.00284	-0.00303	-0.0035	-0.00428	-0.00439	-0.0057	-0.0064
Share EU Subs.	-0.00762	-0.00146	0.00236	0.00134	-0.00158	-0.00958	-0.00899
	-0.00527	-0.00625	-0.00775	-0.00795	-0.00836	-0.00997	-0.0115
Share UK Dir.	0.00387	0.001	0.00361	0.00523	0.0122	-0.00206	-0.00879
	-0.00654	-0.0062	-0.00801	-0.00915	-0.0117	-0.0123	-0.0131
Share EU Dir.	-0.0105	-0.0231*	-0.0262*	-0.0237	-0.00547	-0.0264	-0.0318
	-0.0117	-0.0118	-0.0158	-0.0177	-0.0195	-0.023	-0.0258
Market Capitalisation	0.00071	0.001	0.00244**	0.00361***	0.00234	0.00135	0.00198
	-0.000944	-0.00105	-0.00118	-0.00133	-0.00142	-0.00186	-0.00185
No. Subsidiaries	-0.0000803	-0.00138	-0.00324***	-0.00295**	-0.00284*	-0.00378*	-0.00365*
	-0.000915	-0.001	-0.00119	-0.00137	-0.00145	-0.00199	-0.00205
No. Directors	-0.00174	0.00147	0.00444	0.00671	0.00579	0.00395	0.00366
	-0.00253	-0.00301	-0.00402	-0.00448	-0.00444	-0.00562	-0.00611
Constant	-0.00383	0.00204	-0.00989	-0.0391***	-0.0147	-0.018	-0.0362*
	-0.00942	-0.0102	-0.0115	-0.0133	-0.0139	-0.018	-0.0185
Sector FE	YES	YES	YES	YES	YES	YES	YES
Observations	465	465	465	465	465	465	465
R-squared	0.124	0.13	0.134	0.143	0.102	0.164	0.123

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 8: Event 2, Size of Cumulative Abnormal Returns (CARs)

positive effect on the day, while the share of European directors' effect remains negative. The share of UK subsidiaries is the only significant result and seems to be significant at the 1% level in explaining the CARs for up to 4 days. However, from Table 7, we saw that the abnormal returns due to the event probably only lasted one day, so trying to explain the CARs from (0,+1) onwards might be rather ineffective. Another point to note, is the very low R-squared values, suggesting that this model is not a very good fit and that we are likely omitting a lot of variables. Thus, we are likely suffering from omitted variable bias and so we might be incorrectly concluding that the share of UK subsidiaries helps to explain a firm's CARs for this event.

For the third event, the share of UK subsidiaries and the share of EU subsidiaries now seem to have a positive effect on a firm's CARs (see Table 9). This is because the third event can be seen as a much more "positive" Brexit event than the other two, with the Withdrawal Agreement approved and an end to the years of uncertainty about when and how the UK would leave the EU. Thus, there was more clarity about the future of firms more exposed to the UK and EU, and so investors would likely be more willing to purchase their shares. The nationality of the directors still seems to have a slight negative effect. On the day of the event, the share of UK subsidiaries is significant at the 1% significance level, while the share

VARIABLES	(1) CAR(0,0)	(2) CAR(0,+1)	(3) CAR(0,+2)	(4) CAR(0,+3)	(5) CAR(0,+4)
Share UK Subs.	0.0140***	0.0207***	0.0133***	0.00575	0.00175
	-0.00272	-0.00389	-0.00464	-0.0053	-0.00592
Share EU Subs.	0.00797*	0.0132**	0.00812	-0.00819	-0.00385
	-0.0048	-0.00529	-0.0068	-0.00927	-0.0108
Share UK Dir.	-0.00856	-0.0124	0.00401	-0.00472	-0.00372
	-0.00533	-0.00789	-0.00838	-0.0099	-0.0115
Share EU Dir.	-0.0206*	-0.0115	-0.00515	0.00585	0.00556
	-0.0115	-0.0143	-0.0164	-0.0216	-0.0227
Market Capitalisation	0.00209***	0.00136	0.00408***	0.00273	0.00114
	-0.000786	-0.00115	-0.00153	-0.00199	-0.00206
No. Subsidiaries	-0.000157	-0.000777	-0.00146	-0.00243	-0.00141
	-0.001	-0.00135	-0.0017	-0.00223	-0.00241
No. Directors	-0.00232	0.000531	-0.00546	-0.00873*	-0.00900*
	-0.00277	-0.00335	-0.00456	-0.00506	-0.00499
Constant	-0.0335***	-0.0255**	-0.0672***	-0.0353*	-0.00174
	-0.00823	-0.0117	-0.0157	-0.0213	-0.0214
Sector FE	YES	YES	YES	YES	YES
Observations	473	473	473	473	473
R-squared	0.161	0.175	0.142	0.085	0.069

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 9: Event 3, Size of Cumulative Abnormal Returns (CARs)

of EU subsidiaries and share of European directors are significant at the 10% level, with the share of British directors being insignificant. The following day, the subsidiary structure of a firm remains significant, while the nationality of its directors becomes completely insignificant. From Table 7, we learnt that after the first two days, the abnormal returns are likely not due to the event itself and so again, trying to explain the CARs from then on may be ineffective. Furthermore, similar to the second event, the low R-squared values indicate that this model is not a great fit and so we may be incorrectly concluding that a firm's individual characteristics helps explain its CARs.

4.2.3 Summary/Implications

In summary, the second and third events were far less significant than the first event. This is because they were far more expected than the first event, so if markets are efficient these expectations should be included in the price – the UK was expected to be granted an extension at some point for the second event and the Withdrawal Agreement was expected to be approved by the EU for the third event. There was a spike in the number of statistically significant abnormal returns, but this returned to the pre-event level within a day and within two days for the second and third events respectively. The share of UK subsidiaries seems to play a role in explaining the size of these abnormal returns for the first few days following the two events. However, the low R-squared values suggest that a lot of variables have been omitted and thus makes it difficult to make a robust interpretation.

4.3 Robustness Check

To ensure the results for the first event are robust, the event day was moved to the 23rd June and the 22nd June to allow for any anticipatory effects. This did not have much of an impact on the significance of our results, since there was still a dramatic increase in the number of significant CARs from the 24th June onwards.

VARIABLES	(1) CAR(0,0)	(2) CAR(0,+1)	(3) CAR(0,+2)	(4) CAR(0,+3)	(5) CAR(0,+4)
Share UK Subs.	0.00954***	-0.0783***	-0.168***	-0.159***	-0.157***
	(0.00260)	(0.00823)	(0.0132)	(0.0126)	(0.0127)
Share EU Subs.	0.00954*	-0.0572***	-0.141***	-0.122***	-0.122***
	(0.00536)	(0.0153)	(0.0235)	(0.0226)	(0.0221)
Share UK Dir.	0.00406	-0.0338**	-0.0194	-0.0157	-0.0217
	(0.00483)	(0.0140)	(0.0261)	(0.0259)	(0.0281)
Share EU Dir.	0.00126	-0.0248	-0.0291	-0.0421	-0.0537
	(0.00914)	(0.0291)	(0.0518)	(0.0513)	(0.0511)
Market Capitalisation	-0.00114	-0.00589**	-0.0107***	-0.00593	-0.00698*
	(0.000801)	(0.00246)	(0.00409)	(0.00394)	(0.00394)
No. Subsidiaries	5.21e-05	0.00114	0.00726	0.00507	0.00296
	(0.000947)	(0.00267)	(0.00477)	(0.00460)	(0.00478)
No. Directors	-0.00123	-0.00464	0.000727	-0.00179	8.87e-05
	(0.00214)	(0.00833)	(0.0130)	(0.0133)	(0.0127)
Constant	0.00702	0.144***	0.191***	0.133***	0.159***
	(0.0105)	(0.0315)	(0.0535)	(0.0507)	(0.0523)
Sector FE	YES	YES	YES	YES	YES
Observations	407	407	407	407	407
R-squared	0.195	0.534	0.583	0.549	0.522

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 10: Size of Cumulative Abnormal Returns (CARs) with Event Day on 23rd June

Table 10 shows the results from the second stage regression when the event day is moved to the 23rd June. On the 23rd June, the share of UK subsidiaries and the share of EU subsidiaries seemed to have a slight positive effect on the size of the firm's abnormal returns, with the share of UK subsidiaries being significant at the 1% level and the share of EU subsidiaries being significant at the 10% level. This is probably because on the 23rd June, it was generally believed that the outcome of the referendum would be "Remain", which was good news for firms more exposed to the UK and EU. From the 24th June onwards, the subsidiary structure of a firm had a much larger (negative) effect on the size of its CARs and again, both the share of UK and EU subsidiaries remained significant for 25 days after the event day. The share of British directors also had a significantly negative effect on the CARs from the 23rd June to 24th June, although this time only at the 5% level, compared to the 1% level when the event day was on the 24th. Furthermore, the R-squared jumped from 0.195 on the 23rd June to 0.534 when it includes the 24th June, suggesting that the model is a much better fit after the referendum result was announced.

5 Conclusion

This paper highlights that unexpected events have a much more significant impact on stock markets than expected events. The reaction of the stock market was much more severe to the announcement that the UK would leave the EU than to the UK actually leaving the EU. Other unexpected shocks, such as the Covid-19 pandemic, have caused huge crashes in global stock markets, with the pandemic causing one of the largest stock market crashes since the Great Depression in 1929. Investors swiftly shift their expectations following an unexpected shock so that a firm's actual return may be different from its expected return. However, it seems that even 5 weeks after it was revealed that the United Kingdom would leave the European Union, many firms were still experiencing these abnormal returns, perhaps implying that UK financial markets are slightly inefficient. This paper supports the findings of Ramiah et al. (2016) by showing that a firm's sector played a significant role in explaining the size of its abnormal returns. It further supports Davies & Studnicka (2018) by highlighting that, with a larger sample size, a firm's GVC structure remained significant in explaining the size of its abnormal returns for up to 5 weeks after the result of the referendum was announced. This is also the first paper that attempts to use the nationality of its board of directors to explain the size of a firm's abnormal returns. However, the share of British directors and the share of European directors were both relatively insignificant. Overall, it seems that if another event occurs where trade barriers are increased between two regions, one might expect the firms most exposed to those regions to be more negatively affected.

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A Appendix

A.1 Tables

Sector	*No. of Firms*	*CAAR (0,0)*	*CAAR (0,+1)*	*CAAR (0,+2)*	*CAAR (0,+3)*	*CAAR (0,+4)*
Banking, Insurance & Financial Services	185	-0.75%	-2.72%	-2.45%	-1.96%	-2.45%
Biotechnology and Life Sciences	3	-1.30%	-1.60%	1.72%	3.08%	4.07%
Business Services	46	-4.33%***	-9.89%***	-8.56%***	-8.21%***	-7.95%***
Chemicals, Petroleum, Rubber & Plastic	18	0.13%	-2.23%	-1.54%	-0.76%	0.18%
Communications	7	-3.21%	-3.16%	-3.28%	-2.08%	-0.90%
Computer Software	5	-4.55%*	-10.73%**	-10.03%**	-8.86%***	-8.97%**
Construction	34	-14.10%***	-24.12%***	-21.86%***	-19.29%***	-19.56%***
Food, Beverage & Tobacco Manufacturing	14	-0.81%	-1.64%	-1.13%	0.32%	1.21%
Industrial, Electric & Electronic Machinery	9	-0.71%	-1.99%	-2.47%	-2.94%	-2.67%
Leather, Stone, Clay & Glass products	3	-8.90%	-17.58%	-15.44%	-13.29%	-14.13%
Media & Broadcasting	5	-9.90%*	-13.43%*	-12.30%**	-11.08%**	-11.27%**
Metals & Metal Products	7	-2.12%	-3.92%	-4.27%	-5.51%	-5.79%
Mining & Extraction	22	4.34%	7.46%	5.07%	5.08%	4.59%
Printing & Publishing	9	-1.20%	-4.51%	-5.41%	-5.07%	-5.59%
Property Services	20	-6.45%***	-14.34%***	-12.04%***	-11.24%***	-10.77%***
Public Administration, Education, Health Social Services	8	-5.44%***	-9.80%***	-8.80%***	-9.11%***	-8.91%***
Retail	22	-6.40%***	-15.02%***	-12.76%***	-11.82%***	-12.05%***
Textiles & Clothing Manufacturing	2	1.92%	3.81%**	5.07%***	5.76%***	3.71%***
Transport Manufacturing	5	-1.61%	-2.42%*	-1.51%	-0.94%	-1.00%
Transport, Freight & Storage	18	-5.59%*	-11.52%**	-10.86%**	-9.92%*	-8.57%
Travel, Personal & Leisure	20	-3.58%*	-8.42%**	-7.79%**	-7.73%**	-8.27%**
Utilities	7	-0.62%	0.76%	0.98%	2.36%	4.56%*
Wholesale	17	-3.39%	-7.47%*	-6.05%	-5.00%	-5.57%
Wood, Furniture & Paper Manufacturing	2	2.95%**	4.18%	3.25%	3.34%	4.68%

*** p-value <0.01, ** p-value < 0.05, * p-value < 0.1

Table A.1.1: Event 1, Cumulative Average Abnormal Returns (CAARs) by Sector

Sector	*No. of Firms*	*CAAR (0,+7)*	*CAAR (0,+9)*	*CAAR (0,+14)*	*CAAR (0,+19)*	*CAAR (0,+24)*
Banking, Insurance & Financial Services	185	-3.36%	-2.83%	-0.17%	1.01%	1.77%
Biotechnology and Life Sciences	3	2.22%	1.40%	0.52%	4.04%	1.69%
Business Services	46	-9.83%***	-7.91%**	-3.84%	-1.47%	-0.63%
Chemicals, Petroleum, Rubber & Plastic	18	-0.49%	0.47%	1.58%	2.46%	3.81%
Communications	7	-1.91%	-2.48%	-1.13%	-0.51%	1.45%
Computer Software	5	-9.89%**	-9.28%**	-2.69%	3.55%	7.93%
Construction	34	-27.10%***	-27.47%***	-19.05%***	-17.91%***	-16.01%***
Food, Beverage & Tobacco Manufacturing	14	3.06%	4.87%	4.23%	5.14%	5.84%
Industrial, Electric & Electronic Machinery	9	-4.35%	-3.48%	0.43%	0.27%	3.48%
Leather, Stone, Clay & Glass products	3	-19.62%	-22.81%	-15.93%*	-11.35%	-10.91%
Media & Broadcasting	5	-11.60%*	-10.05%*	-6.06%	-4.73%	-3.39%
Metals & Metal Products	7	-8.20%	-8.33%*	-3.50%	-3.20%	-1.73%
Mining & Extraction	22	6.79%	7.13%	8.81%	4.62%	3.52%
Printing & Publishing	9	-6.89%	-7.02%	-4.32%	-1.09%	0.73%
Property Services	20	-15.82%***	-16.35%***	-9.51%***	-7.81%***	-5.93%**
Public Administration, Education, Health Social Services	8	-11.52%***	-10.76%***	-8.43%***	-8.27%***	-8.71%***
Retail	22	-16.36%***	-16.34%***	-10.11%***	-9.18%***	-9.05%***
Textiles & Clothing Manufacturing	2	4.17%***	2.46%	5.92%	7.84%	12.77%***
Transport Manufacturing	5	-2.88%	-1.69%	0.29%	1.01%	5.25%
Transport, Freight & Storage	18	-12.86%**	-14.20%**	-10.83%**	-12.24%**	-10.71%*
Travel, Personal & Leisure	20	-8.72%**	-7.15%*	-3.14%	-0.73%	0.89%
Utilities	7	3.46%	2.66%	2.83%	3.28%**	3.64%**
Wholesale	17	-6.87%	-5.81%	-4.17%	-3.04%	-3.07%
Wood, Furniture & Paper Manufacturing	2	5.18%	3.95%	2.38%	2.52%***	4.34%***

*** p-value <0.01, ** p-value <0.05, * p-value <0.1

Table A.1.2: Event 1, Long-Term Cumulative Average Abnormal Returns (CAARs) by Sector

Sector	*No. of Firms*	*CAAR (0,0)*	*CAAR (0,+1)*	*CAAR (0,+2)*	*CAAR (0,+3)*	*CAAR (0,+4)*
Banking, Insurance & Financial Services	206	-0.31%	-0.38%	-0.67%	-0.89%*	-0.67%
Biotechnology and Life Sciences	3	-1.28%	-0.91%	-2.80%	-4.20%***	-3.96%*
Business Services	65	-0.85%	-0.71%	-1.31%*	-1.06%	-0.87%
Chemicals, Petroleum, Rubber & Plastic	18	-0.45%	-0.38%	-1.29%	-1.12%	-1.50%
Communications	7	-0.56%	-1.29%	-2.00%**	-1.99%	-1.93%
Computer Software	7	0.46%	1.66%**	-0.69%	-1.69%	-1.58%
Construction	34	-1.76%	-1.56%	-1.81%	-1.84%	-1.12%
Food, Beverage & Tobacco Manufacturing	14	-0.64%	-0.85%	-1.21%	-1.47%	-1.31%*
Industrial, Electric & Electronic Machinery	10	-1.54%	-0.93%	-1.94%	-2.35%*	-2.61%
Leather, Stone, Clay & Glass products	4	-0.46%	-0.38%	-0.21%	-2.66%	-1.85%
Media & Broadcasting	5	-0.55%	0.07%	-1.14%	-1.07%	-0.23%
Metals & Metal Products	7	0.22%	0.20%	-0.59%	-0.93%	-1.04%
Mining & Extraction	24	0.13%	0.45%	0.83%	1.31%	1.06%
Printing & Publishing	9	0.00%	0.02%	-1.22%	-1.49%	-1.11%
Property Services	21	-0.23%	-0.78%*	-0.61%	-0.65%	0.33%
Public Administration, Education, Health Social Services	9	-1.16%**	-1.67%**	-2.46%***	-1.75%*	-1.44%
Retail	22	-0.95%	-0.07%	-0.50%	-0.45%	-0.10%
Textiles & Clothing Manufacturing	2	-0.78%**	-0.10%	-0.73%	-0.67%	-0.81%
Transport Manufacturing	5	-1.12%	-1.14%	-2.51%	-2.89%	-2.24%*
Transport, Freight & Storage	18	-0.59%	-1.22%	-2.09%	-2.50%*	-2.59%*
Travel, Personal & Leisure	26	-0.21%	-0.17%	-0.81%	-1.00%	-0.29%
Utilities	7	-0.96%	-0.70%	-1.93%	-1.84%	-1.91%
Wholesale	18	0.04%	0.84%*	0.30%	0.12%	0.57%
Wood, Furniture & Paper Manufacturing	2	1.24%	0.87%	-0.25%	-0.22%	-0.40%

*** p-value <0.01, ** p-value <0.05, * p-value <0.1

Table A.1.3: Event 2, Cumulative Average Abnormal Returns (CAARs) by Sector

Sector	*No. of Firms*	*CAAR (0,+7)*	*CAAR (0,+9)*	*CAAR (0,+14)*	*CAAR (0,+19)*	*CAAR (0,+24)*
Banking, Insurance & Financial Services	206	-0.17%	0.15%	0.22%	1.12%	1.80%
Biotechnology and Life Sciences	3	-5.57%***	-4.65%*	-3.47%	1.48%	2.06%
Business Services	65	-0.15%	0.81%	1.33%	3.46%	4.83%*
Chemicals, Petroleum, Rubber & Plastic	18	-0.54%	-0.30%	-3.57%	-1.17%	0.34%
Communications	7	-2.60%	-2.01%	-2.80%*	-2.95%	-1.52%
Computer Software	7	-2.03%	-2.18%	-4.37%	-4.61%	0.16%
Construction	34	-2.76%*	-1.86%	-3.41%	-2.88%	-2.93%*
Food, Beverage & Tobacco Manufacturing	14	-1.47%*	-2.12%***	-1.73%**	-1.80%*	-0.72%
Industrial, Electric & Electronic Machinery	10	-1.61%	-0.37%	0.85%	5.10%*	4.14%
Leather, Stone, Clay & Glass products	4	-0.77%	0.91%	-1.91%	-0.21%	1.68%
Media & Broadcasting	5	0.23%	1.98%	1.83%	5.24%	5.41%*
Metals & Metal Products	7	0.04%	1.12%	1.64%	4.86%**	3.41%
Mining & Extraction	24	2.99%	1.98%	2.72%	0.10%	-1.38%
Printing & Publishing	9	-0.20%	1.33%	0.22%	1.75%	4.62%*
Property Services	21	1.61%	2.31%	1.22%	0.90%	0.44%
Public Administration, Education, Health Social Services	9	-1.84%	-2.62%	-11.05%	-9.99%	-7.24%
Retail	22	0.65%	0.52%	0.05%	2.54%	4.69%
Textiles & Clothing Manufacturing	2	-0.41%	-1.89%	-2.50%	-0.31%	2.12%
Transport Manufacturing	5	0.31%	0.64%	-0.95%	1.26%	2.28%
Transport, Freight & Storage	18	-4.23%*	-2.54%	-4.69%	-2.97%	-1.22%
Travel, Personal & Leisure	26	-0.23%	-0.63%	-0.57%	2.06%	5.71%
Utilities	7	-3.78%	-3.49%	-3.98%	-5.65%	-3.80%
Wholesale	18	0.75%	0.91%	1.40%	1.39%	2.73%
Wood, Furniture & Paper Manufacturing	2	0.78%	-0.27%	-2.58%	-7.21%***	-4.01%***

*** p-value <0.01, ** p-value <0.05, * p-value <0.1

Table A.1.4: Event 2, Long-Term Cumulative Average Abnormal Returns (CAARs) by Sector

VARIABLES	(8) CAR(0,+14)	(9) CAR(0,+19)	(10) CAR(0,+24)
Share UK Subs.	-0.0231*	-1.70e-05	0.00371
	(0.0127)	(0.0129)	(0.0136)
Share EU Subs.	-0.0331	-0.00503	-0.00298
	(0.0464)	(0.0392)	(0.0403)
Share UK Dir.	0.0319	0.0201	0.0145
	(0.0708)	(0.0588)	(0.0585)
Share EU Dir.	0.0467	0.0516	0.0559
	(0.114)	(0.0949)	(0.0952)
Market Capitalisation	0.00300	0.000410	-0.00219
	(0.00461)	(0.00449)	(0.00484)
No. Subsidiaries	-0.00130	-0.00125	-0.000874
	(0.00315)	(0.00377)	(0.00412)
No. Directors	-0.00787	-0.00675	-0.0144
	(0.0130)	(0.0136)	(0.0146)
Constant	-0.0118	0.0449	0.0762*
	(0.0442)	(0.0421)	(0.0461)
Sector FE	YES	YES	YES
Observations	465	465	465
R-squared	0.113	0.123	0.118

*** p-value <0.01, ** p-value <0.05, * p-value <0.1

Table A.1.5: Event 2, Long-Term Size of Cumulative Abnormal Returns (CARs)

Sector	*No. of Firms*	*CAAR (0,0)*	*CAAR (0,+1)*	*CAAR (0,+2)*	*CAAR (0,+3)*	*CAAR (0,+4)*
Banking, Insurance & Financial Services	208	-0.22%	-0.11%	-0.10%	0.30%	0.55%
Biotechnology and Life Sciences	3	2.13%**	1.94%	-1.46%	-0.14%	0.32%
Business Services	69	0.01%	0.56%	0.32%	0.24%	0.40%
Chemicals, Petroleum, Rubber & Plastic	18	-0.06%	0.23%	0.12%	0.46%	0.91%
Communications	7	-2.01%**	-2.20%**	-1.56%	-1.94%**	-2.70%*
Computer Software	7	-1.11%	-1.27%	-1.74%	-1.54%	-1.91%
Construction	34	1.19%	2.04%	2.11%	2.45%	1.71%
Food, Beverage & Tobacco Manufacturing	13	-0.24%	0.91%	0.33%	0.22%	-0.39%
Industrial, Electric & Electronic Machinery	10	-0.04%	0.22%	-0.07%	-1.17%	-0.46%
Leather, Stone, Clay & Glass products	4	0.84%***	1.47%	-0.88%	-0.26%	-0.17%
Media & Broadcasting	4	-0.69%	-0.79%	-0.87%	-1.12%	-1.47%
Metals & Metal Products	7	0.47%	1.17%*	0.11%	2.34%	3.84%***
Mining & Extraction	24	0.57%	0.42%	-1.25%	-1.39%	-0.39%
Printing & Publishing	9	0.60%	-1.27%	-0.23%	-0.54%	-0.94%
Property Services	20	0.14%	0.48%	-0.31%	-0.10%	0.33%
Public Administration, Education, Health Social Services	9	0.26%	0.47%	0.04%	0.34%	-0.37%
Retail	23	0.59%	0.38%	1.21%	1.08%	1.29%
Textiles & Clothing Manufacturing	2	1.10%***	0.81%	0.22%	0.38%***	-1.98%
Transport Manufacturing	5	0.75%**	1.74%***	2.42%***	1.33%	0.25%
Transport, Freight & Storage	18	-0.09%	0.77%	0.01%	-0.28%	-0.35%
Travel, Personal & Leisure	25	0.48%	0.89%	0.91%	0.55%	0.72%
Utilities	7	0.63%	0.93%	1.01%	-0.57%	-1.10%
Wholesale	18	0.27%	0.69%	0.48%	1.47%	0.90%
Wood, Furniture & Paper Manufacturing	2	0.79%*	0.35%	1.18%	0.60%	2.87%***

*** p-value <0.01, ** p-value <0.05, * p-value <0.1

Table A.1.6: Event 3, Cumulative Average Abnormal Returns (CAARs) by Sector

Sector	*No. of Firms*	*CAAR (0,+7)*	*CAAR (0,+9)*	*CAAR (0,+14)*
Banking, Insurance & Financial Services	208	0.47%	0.65%	0.96%
Biotechnology and Life Sciences	3	1.13%	-0.27%	1.82%
Business Services	69	0.41%	0.49%	0.71%
Chemicals, Petroleum, Rubber & Plastic	18	0.87%	1.38%	-0.53%
Communications	7	-3.12%*	-2.12%	-0.77%
Computer Software	7	-0.57%	-1.85%	-2.81%
Construction	34	1.69%	1.88%	4.27%
Food, Beverage & Tobacco Manufacturing	13	0.59%	0.32%	0.46%
Industrial, Electric & Electronic Machinery	10	-3.48%	-2.49%	-3.12%
Leather, Stone, Clay & Glass products	4	0.07%	1.29%	0.43%
Media & Broadcasting	4	-1.84%	-0.87%	-2.23%
Metals & Metal Products	7	3.02%	4.29%**	3.29%
Mining & Extraction	24	-2.65%	-1.16%	-0.33%
Printing & Publishing	9	-1.35%	-0.45%	-1.82%
Property Services	20	0.00%	-0.72%	0.51%
Public Administration, Education, Health Social Services	9	-0.95%	-1.35%	-1.69%
Retail	23	1.35%	1.39%	2.34%
Textiles & Clothing Manufacturing	2	-2.76%**	-2.77%	-2.08%***
Transport Manufacturing	5	1.24%	1.18%	-3.04%
Transport, Freight & Storage	18	-1.21%	-0.88%	-0.70%
Travel, Personal & Leisure	25	-0.17%	1.27%	2.01%
Utilities	7	-2.19%	-1.34%	-1.08%
Wholesale	18	0.85%	0.19%	0.51%
Wood, Furniture & Paper Manufacturing	2	3.71%***	3.78%**	3.95%***

*** p-value <0.01, ** p-value <0.05, * p-value <0.1

Table A.1.7: Event 3, Long-Term Cumulative Average Abnormal Returns (CAARs) by Sector

VARIABLES	(6) CAR(0,+7)	(7) CAR(0,+9)	(8) CAR(0,+14)
Share UK Subs.	-0.00525	-0.00350	0.000699
	(0.00716)	(0.00757)	(0.0101)
Share EU Subs.	-0.0130	-0.00778	-0.0124
	(0.0150)	(0.0164)	(0.0181)
Share UK Dir.	-0.00336	-0.00714	0.00885
	(0.0195)	(0.0220)	(0.0210)
Share EU Dir.	0.00434	-0.00550	0.0158
	(0.0333)	(0.0371)	(0.0414)
Market Capitalisation	0.00344	0.00146	0.00644*
	(0.00242)	(0.00287)	(0.00388)
No. Subsidiaries	-0.00183	-0.00139	-0.00401
	(0.00269)	(0.00310)	(0.00422)
No. Directors	-0.00903	-0.00408	-0.0109
	(0.00717)	(0.00731)	(0.0108)
Constant	0.0481*	0.0298	-0.0305
	(0.0274)	(0.0316)	(0.0438)
Sector FE	YES	YES	YES
Observations	473	473	473
R-squared	0.092	0.064	0.074

*** p-value <0.01, ** p-value <0.05, * p-value <0.1

Table A.1.8: Event 3, Long-Term Size of Cumulative Abnormal Returns (CARs)

Window	*Observations*	*Mean (%)*	*St. Dev. (%)*	*Min (%)*	*Max (%)*	*No. Significant*
(-1,-1)	488	0.05	1.61	-8.57	9.59	57
(0,0)	488	0.41	1.62	-5.42	10.13	54
(0,+1)	488	-2.67	6.66	-30.04	15.04	248
(0,+2)	488	-6.34	11.71	-61.59	22.92	330
(0,+3)	488	-5.74	10.93	-61.10	18.51	304
(0,+4)	488	-5.14	10.66	-53.74	25.05	282
(0,+7)	488	-5.42	12.53	-51.82	40.42	262
(0,+9)	488	-7.35	15.29	-65.34	46.44	286
(0,+14)	488	-3.89	12.30	-43.94	38.83	206
(0,+19)	488	-3.00	11.98	-47.49	42.89	154
(0,+24)	488	-2.15	12.34	-52.78	43.40	141

Table A.1.9: Cumulative Abnormal Returns (CARs) with Event Day on 23rd June

VARIABLES	(6) CAR(0,+7)	(7) CAR(0,+9)	(8) CAR(0,+14)	(9) CAR(0,+19)	(10) CAR(0,+24)
Share UK Subs.	-0.229***	-0.251***	-0.209***	-0.197***	-0.201***
	(0.0168)	(0.0175)	(0.0161)	(0.0163)	(0.0180)
Share EU Subs.	-0.166***	-0.163***	-0.120***	-0.0962***	-0.0935***
	(0.0272)	(0.0292)	(0.0276)	(0.0284)	(0.0305)
Share UK Dir.	-0.0209	-0.0161	-0.0175	-0.0301	-0.0212
	(0.0358)	(0.0394)	(0.0342)	(0.0377)	(0.0431)
Share EU Dir.	-0.00185	-0.0195	-0.00220	-0.00507	-0.0250
	(0.0625)	(0.0629)	(0.0542)	(0.0586)	(0.0653)
Market Capitalisation	-0.0106**	-0.00921*	-0.0106**	-0.0143***	-0.0144**
	(0.00517)	(0.00554)	(0.00492)	(0.00501)	(0.00557)
No. Subsidiaries	0.00210	0.000333	0.000580	-0.000327	-0.00283
	(0.00612)	(0.00656)	(0.00596)	(0.00604)	(0.00642)
No. Directors	0.00550	0.00824	0.0113	0.0115	0.0144
	(0.0163)	(0.0173)	(0.0161)	(0.0161)	(0.0166)
Constant	0.218***	0.213***	0.228***	0.295***	0.301***
	(0.0669)	(0.0689)	(0.0569)	(0.0658)	(0.0734)
Sector FE	YES	YES	YES	YES	YES
Observations	407	407	407	407	407
R-squared	0.562	0.573	0.546	0.519	0.474

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table A.1.10: Long-Term Size of Cumulative Abnormal Returns (CARs) with Event Day on 23rd June

Window	*Observations*	*Mean (%)*	*St. Dev. (%)*	*Min (%)*	*Max (%)*	*No. Significant*
(-1,-1)	488	-1.11	4.12	-20.79	17.98	220
(0,0)	488	0.05	1.61	-8.57	9.59	57
(0,+1)	488	0.46	2.18	-10.72	8.63	59
(0,+2)	488	-2.62	6.52	-29.10	14.25	207
(0,+3)	488	-6.29	11.45	-57.85	22.11	308
(0,+4)	488	-5.69	10.65	-57.36	17.71	288
(0,+7)	488	-4.71	11.06	-45.76	31.84	251
(0,+9)	488	-7.17	13.78	-60.34	39.30	264
(0,+14)	488	-3.94	11.85	-42.75	37.85	203
(0,+19)	488	-3.22	11.96	-47.52	43.32	158
(0,+24)	488	-2.71	12.76	-49.00	46.48	133

Table A.1.11: Cumulative Abnormal Returns (CARs) with Event Day on 22nd June

VARIABLES	(1) CAR(0,0)	(2) CAR(0,+1)	(3) CAR(0,+2)	(4) CAR(0,+3)	(5) CAR(0,+4)
Share UK Subs.	0.0115***	0.0210***	-0.0668***	-0.156***	-0.148***
	(0.00240)	(0.00359)	(0.00846)	(0.0131)	(0.0123)
Share EU Subs.	0.00231	0.0118*	-0.0549***	-0.139***	-0.120***
	(0.00497)	(0.00682)	(0.0157)	(0.0237)	(0.0226)
Share UK Dir.	-0.00562	-0.00156	-0.0394***	-0.0250	-0.0213
	(0.00580)	(0.00641)	(0.0152)	(0.0267)	(0.0265)
Share EU Dir.	-0.00481	-0.00355	-0.0296	-0.0339	-0.0469
	(0.00914)	(0.0116)	(0.0298)	(0.0520)	(0.0509)
Market Capitalisation	0.00110	-4.21e-05	-0.00479*	-0.00964**	-0.00483
	(0.000853)	(0.00111)	(0.00257)	(0.00414)	(0.00396)
No. Subsidiaries	-0.00115	-0.00109	-1.40e-06	0.00612	0.00392
	(0.000797)	(0.00125)	(0.00284)	(0.00484)	(0.00463)
No. Directors	0.00502**	0.00379	0.000381	0.00575	0.00323
	(0.00251)	(0.00350)	(0.00856)	(0.0130)	(0.0130)
Constant	-0.0164	-0.00937	0.127***	0.175***	0.116**
	(0.0101)	(0.0125)	(0.0324)	(0.0541)	(0.0510)
Sector FE	YES	YES	YES	YES	YES
Observations	407	407	407	407	407
R-squared	0.166	0.201	0.481	0.561	0.527

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table A.1.12: Size of Cumulative Abnormal Returns (CARs) with Event Day on 22nd June

VARIABLES	(6) CAR(0,+7)	(7) CAR(0,+9)	(8) CAR(0,+14)	(9) CAR(0,+19)	(10) CAR(0,+24)
Share UK Subs.	-0.157***	-0.208***	-0.190***	-0.181***	-0.186***
	(0.0137)	(0.0161)	(0.0156)	(0.0162)	(0.0176)
Share EU Subs.	-0.132***	-0.154***	-0.136***	-0.0944***	-0.0751**
	(0.0243)	(0.0270)	(0.0278)	(0.0299)	(0.0304)
Share UK Dir.	-0.0203	-0.0224	-0.0101	-0.0208	-0.0319
	(0.0311)	(0.0357)	(0.0357)	(0.0364)	(0.0406)
Share EU Dir.	-0.0253	-0.00540	0.00459	-0.00854	-0.0359
	(0.0546)	(0.0612)	(0.0567)	(0.0561)	(0.0629)
Market Capitalisation	-0.00743*	-0.0106**	-0.00888*	-0.0113**	-0.0140**
	(0.00434)	(0.00510)	(0.00460)	(0.00489)	(0.00549)
No. Subsidiaries	0.00175	0.00101	-0.000848	-0.00136	-0.00182
	(0.00529)	(0.00604)	(0.00573)	(0.00603)	(0.00660)
No. Directors	0.00317	0.00929	0.0175	0.0128	0.0199
	(0.0134)	(0.0158)	(0.0144)	(0.0158)	(0.0167)
Constant	0.164***	0.208***	0.184***	0.235***	0.281***
	(0.0553)	(0.0662)	(0.0587)	(0.0615)	(0.0701)
Sector FE	YES	YES	YES	YES	YES
Observations	407	407	407	407	407
R-squared	0.470	0.535	0.502	0.483	0.466

Robust standard errors in parentheses
*** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table A.1.13: Long-Term Size of Cumulative Abnormal Returns (CARs) with Event Day on 22nd June